Case name: Test 1

Description:

Tue Nov 10 14:15:22 2026

Table 1. Propellant Specification

Component	Temperature, K	Pressure, MPa	Enthalpy, kJ/ mol	Enthalpy, kJ/ kg	Mass fraction
RP-1	298.1	0.1013	-24.7177	-1768.557	0.4000000
O2(L)	90.2	0.1013	-12.9790	-405.608	0.6000000
Total			-20.0712	-950.788	

Propellant exploded formula:

$C_{0.604172}\ H_{1.178135}\ O_{0.791656}$ (based on 1 mole)

$C_{23.472436}\ H_{45.771249}\ O_{30.756315}$ (by mass %)

α:	0.4404415	(oxidizer excess coefficient)
O/F:	1.5000000	
O/F_0:	3.4056740	(stoichiometric)
rho:	941.03231	kg/m^3

Table 2. Combustion Properties

Parameter	Injector	Nozzle inlet	Nozzle throat	Nozzle exit	Unit
Pressure	2.0000	1.9759	1.0985	0.0089	MPa
Temperature	2489.8289	2486.8058	2209.6686	877.2925	K
Enthalpy	-16791.3801	-16917.5649	-28398.9090	-84996.9683	J/mol
	-950.7900	-957.9224	-1606.1601	-4736.8389	kJ/kg
Entropy	227.8879	227.9410	228.2064	231.5953	J/(mol·K)
	12.9039	12.9067	12.9067	12.9067	kJ/(kg·K)
Internal energy	-37492.9928	-37594.0423	-46771.1363	-92191.9289	J/mol
	-2122.9919	-2128.6856	-2645.2401	-5137.8105	kJ/kg
Specific heat (p=const)	2.4520	2.4506	2.2984	20.2902	kJ/(kg·K)
Specific heat (V=const)	1.9630	1.9617	1.8215	16.8308	kJ/(kg·K)
Gamma	1.2491	1.2492	1.2618	1.2055	
Isentropic exponent	1.2480	1.2482	1.2614	1.1060	
Gas constant	0.4708	0.4708	0.4702	0.4571	kJ/(kg·K)
Molecular weight	17.6605	17.6607	17.6812	17.9438	
Density	1.7062	1.6877	1.0571	0.0223	kg/m^3
Sonic velocity	1209.5300	1208.8409	1144.8751	665.9263	m/s
Mach number	0.0000	0.0988	1.0000	4.1322	
Area ratio	6.0000	6.0000	1.0000	19.7400	A/At
Mass flux	201.5741	201.5741	1210.2921	61.2860	kg/(m^2·s)
Viscosity	0.7701	0.7695	0.7077	0.3769 x10^{-4}	kg/(m·s)

Parameter	Injector	Nozzle inlet	Nozzle throat	Nozzle exit	Unit
Conductivity, frozen	0.3272	0.3268	0.2942	0.1456	W/(m·K)
Specific heat (p=const), frozen	2.2303	2.2299	2.1937	1.9141	kJ/(kg·K)
Prandtl number, frozen	0.5249	0.5250	0.5277	0.4955	
Conductivity, effective	0.4155	0.4146	0.3291	0.1456	W/(m·K)
Specific heat (p=const), effective	2.4520	2.4506	2.2984	1.9141	kJ/(kg·K)
Prandtl number, effective	0.4545	0.4548	0.4942	0.4955	

Table 3. Combustion Products

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
C(gr)							0.0091088	0.0136084
CH4	0.0000003	0.0000003	0.0000002	0.0000003	0.0000003	0.0000003	0.0063791	0.0071351
CO	0.7578572	0.4778317	0.7578126	0.4778098	0.7528562	0.4752374	0.5752638	0.3685255
CO2	0.0687937	0.0276061	0.0688640	0.0276346	0.0766618	0.0307996	0.3048257	0.1242855
COOH	0.0000007	0.0000003	0.0000007	0.0000003	0.0000002	0.0000001		
H	0.0001849	0.0032398	0.0001836	0.0032163	0.0000620	0.0010884		
H2	0.0413295	0.3620745	0.0413339	0.3621179	0.0418013	0.3666391	0.0483779	0.4306229
H2O	0.1315703	0.1289790	0.1315445	0.1289554	0.1285570	0.1261734	0.0560448	0.0558225
HCHO,formaldehy	0.0000024	0.0000014	0.0000024	0.0000014	0.0000013	0.0000008		
HCO	0.0000073	0.0000044	0.0000072	0.0000044	0.0000021	0.0000013		
HCOOH	0.0000012	0.0000005	0.0000012	0.0000005	0.0000007	0.0000003		
O	0.0000013	0.0000014	0.0000012	0.0000014				
O2	0.0000004	0.0000002	0.0000004	0.0000002				
OH	0.0002508	0.0002604	0.0002480	0.0002576	0.0000569	0.0000592		

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
Gaseous fraction:	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	0.9908912	0.9863916
Condensed fraction:	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0091088	0.0136084

Table 4. Ideal Performance

Parameter	Sea level	Optimum expansion	Vacuum	Units
Characteristic velocity		1642.5709		m/s
Effective exhaust velocity	1244.1465	2751.7445	2897.4598	m/s
Specific impulse (by mass)	1244.1465	2751.7445	2897.4598	N·s/kg
Specific impulse (by weight)	126.8676	280.5998	295.4587	s
Thrust coefficient	0.7574	1.6753	1.7640	
Thrust	28.3811	62.7721	66.0961	kN
Altitude	0.0000	16.9418		km
Ambient pressure	1.0000	0.0881	0.0000	atm

Table 5. Estimated Delivered Performance

Parameter	Sea level	Optimum expansion	Vacuum	Units
Characteristic velocity		1589.7027		m/s
Effective exhaust velocity	1171.8881	2630.9621	2771.9874	m/s
Specific impulse (by mass)	1171.8881	2630.9621	2771.9874	N·s/kg
Specific impulse (by weight)	119.4993	268.2835	282.6640	s
Thrust coefficient	0.7372	1.6550	1.7437	
Thrust	26.7328	60.0168	63.2338	kN
Altitude	0.0000	16.9418		km
Ambient pressure	1.0000	0.0881	0.0000	atm

Table 6. Altitude Performance

Altitude, km	Pressure, atm	Specific impulse, N·s/kg	Specific impulse, s	Thrust coefficient	Thrust, kN
0.0000	1.0000	1171.8881	119.4993	0.7372	26.7328
0.1252	0.9852	1195.5995	121.9172	0.7521	27.2737
0.3164	0.9630	1231.0898	125.5362	0.7744	28.0833
0.5436	0.9372	1272.4361	129.7524	0.8004	29.0265

Altitude, km	Pressure, atm	Specific impulse, N·s/kg	Specific impulse, s	Thrust coefficient	Thrust, kN
0.7979	0.9089	1317.6498	134.3629	0.8289	30.0579
1.0749	0.8790	1365.5653	139.2489	0.8590	31.1509
1.3708	0.8478	1415.3763	144.3282	0.8903	32.2872
1.6837	0.8159	1466.4758	149.5389	0.9225	33.4529
2.0117	0.7835	1518.3833	154.8320	0.9551	34.6370
2.3538	0.7508	1570.7063	160.1675	0.9881	35.8305
2.7090	0.7180	1623.1171	165.5119	1.0210	37.0261
3.0761	0.6854	1675.3384	170.8370	1.0539	38.2174
3.4545	0.6530	1727.1335	176.1186	1.0865	39.3989
3.8436	0.6210	1778.2992	181.3361	1.1186	40.5661
4.2428	0.5895	1828.6609	186.4715	1.1503	41.7149
4.6516	0.5587	1878.0683	191.5097	1.1814	42.8420
5.0694	0.5285	1926.3925	196.4374	1.2118	43.9443
5.4964	0.4990	1973.5236	201.2434	1.2414	45.0195
5.9318	0.4704	2019.3684	205.9183	1.2703	46.0653
6.3750	0.4426	2063.8491	210.4540	1.2983	47.0800
6.8265	0.4157	2106.9015	214.8442	1.3253	48.0621
7.2853	0.3897	2148.4735	219.0833	1.3515	49.0104
7.7513	0.3646	2188.5247	223.1674	1.3767	49.9240
8.2249	0.3406	2227.0248	227.0933	1.4009	50.8023
8.7049	0.3175	2263.9529	230.8589	1.4241	51.6447
9.1918	0.2954	2299.2966	234.4630	1.4464	52.4509
9.6854	0.2743	2333.0515	237.9050	1.4676	53.2209
10.1854	0.2542	2365.2201	241.1853	1.4878	53.9548
10.6915	0.2351	2395.8117	244.3048	1.5071	54.6526
11.2035	0.2170	2424.8134	247.2622	1.5253	55.3142
11.7218	0.2000	2451.9675	250.0311	1.5424	55.9336

Table 7. Throttled Performance

Table 8. Chamber Size

Combustion chamber size			Nozzle size	
Dc	373.30	mm	Type	Parabolic nozzle
Dt	152.40	mm		
Lcyl	781.13	mm	Rn	29.11 mm
Lc	1020.00	mm	Tn	30.00 deg

L*	5407.90	mm	Te	5.00	deg
R1	114.30	mm	De	677.11	mm
R2	63.20	mm	Le	1087.20	mm
b	30.00	deg	Le/Dt	7.13	
Ac/At	6.00		Le/Lc15	110.61	%
			Ae/At	19.74	

Parameter		**Engine**	**Chamber**	
Thrust	sea level	26.7328	26.7328	kN
	opt exp	60.0168	60.0168	kN
	vacuum	63.2338	63.2338	kN
Specific Impulse	sea level	1171.8881	1171.8881	N·s/kg
	opt exp	2630.9621	2630.9621	N·s/kg
	vacuun	2771.9874	2771.9874	N·s/kg
Mass flow rate	total	22.8117	22.8117	kg/s
	oxidizer	13.6870	13.6870	kg/s
	fuel	9.1247	9.1247	kg/s
	film coolant	n.a.	n.a.	kg/s
Number of chambers		1		

Table 10. Thermal Analysis